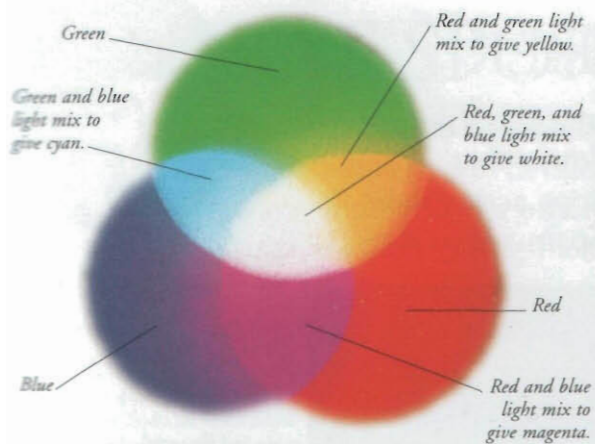




Coloured lights

Different amounts of red, green, and blue light can be mixed to form light of any other colour. This process is called colour addition. Unlike paints, red, green, and blue are the primary colours of light. Equal amounts of any two primary colours give a secondary colour (yellow, cyan, or magenta). When all three primaries are mixed in equal amounts, white light is produced.

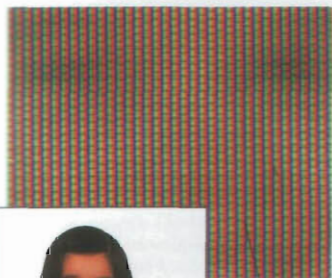


Image is formed by tiny glowing strips.

Colour television

The principle of adding coloured lights is used in colour television. The screen is covered with tiny strips that glow with red, green, or blue light. They are so small that, at a normal viewing distance, the human eye mixes the light coming from them. By adjusting the intensities of these three colours, the sensation of any other colour can be produced.

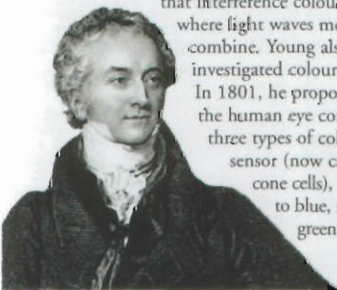
Painting with dots

"Pointillism" is a style of painting in which an artist uses thousands of tiny coloured dots to build up a picture. When viewed close up, the colours of the individual dots are clearly visible. Like the coloured strips on a television screen, the dots are too small to be seen from farther away. When viewed from a distance, the dots seem to merge, giving areas a single colour.



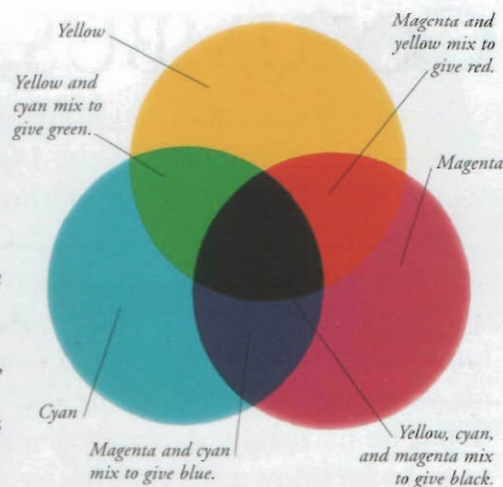
Thomas Young

The English doctor and physicist Thomas Young (1773–1829) carried out many experiments to prove that light travels as waves. He realized that colours are light waves of different lengths, and that interference colours occur where light waves meet and combine. Young also investigated colour vision. In 1801, he proposed that the human eye contains three types of colour sensor (now called cone cells), sensitive to blue, red, and green light.



Pigments

A pigment is a chemical that absorbs only certain colours from white light. This process is called colour subtraction. Yellow, magenta, and cyan are primary pigments. Each absorbs one of the primary colours of light and reflects the other two. For example, a yellow pigment absorbs blue light but reflects green and red, which mix to give yellow. An equal mix of all three pigments absorbs all the colours from white light, giving black.



Colour printing

To print a full-colour picture, three single-colour images are printed on top of each other – one in cyan, one in magenta, and one in yellow. Each picture is made up of tiny coloured dots. The dots overlap and absorb the right wavelengths of light to give all the other colours required. A black image is then added to make the picture sharper.



Picture is made up of tiny ink dots.

Mixing paints

Paints are pigments mixed with water or oil. Any colour except white can be made by mixing the three primary pigments. Mixing paints has the effect of evenly mixing the pigments, and absorbing more of the white light spectrum.



Scattering and interference

Two other processes, called scattering and interference, can remove colours from the spectrum. Interference occurs when light from two sources meets and combines. In scattering, some parts of the spectrum are briefly absorbed by particles of matter and then radiated out again in all directions.



Blue sky

Sunlight includes all the colours of the spectrum. The sky appears blue during the day because air molecules in the atmosphere scatter light from the blue end of the spectrum in all directions.

Soap bubble

When white light strikes a soap bubble, it reflects off both the inner and outer surfaces of the bubble. The reflected light rays interfere, making some colours cancel each other out but others appear bright.

Interference creates a pattern of bright colours and dark bands.



Using interference

Stress is a force that can stretch or bend objects. Engineers shine light through plastic models of their designs to test their ability to withstand stress. The plastic molecules make the light rays split up and interfere. The interference patterns show the points of greatest stress.



Reflecting colours

Objects have colour only when light falls upon them, because colours do not exist in total darkness. An object that appears one colour in white light may look different when illuminated by coloured light. The yellow pot in this sequence of pictures appears yellow only in white light.

White light

The yellow pot reflects the red and green parts of the white light spectrum, but absorbs the blue part.



Red light

The yellow pot reflects red light, and therefore appears red when illuminated by red light.



Green light

When illuminated by green light, the yellow pigment reflects the green light and so appears green.

Blue light
When only blue light is available, the yellow pot absorbs the blue light, making it look black.

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